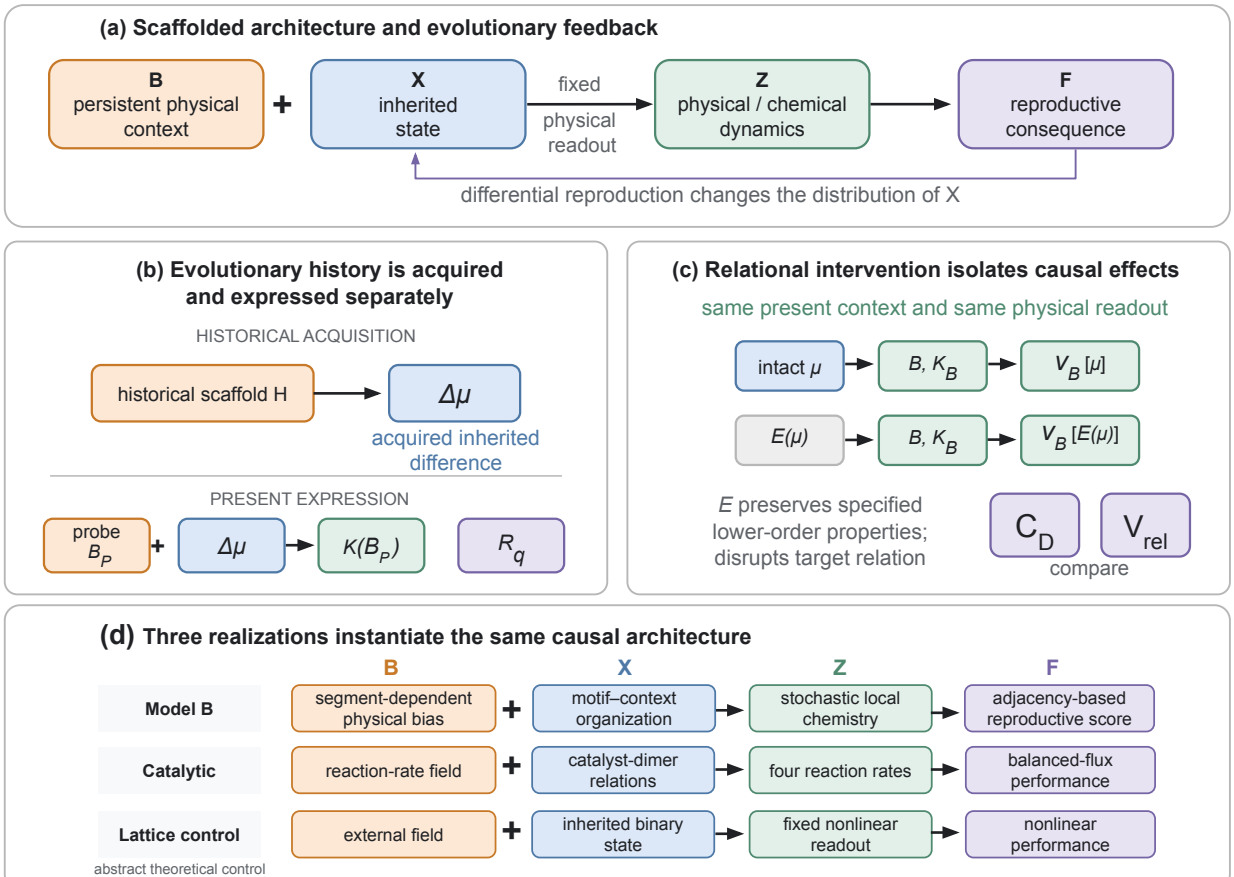

(a) Scaffolded architecture and evolutionary feedback
B
persistent physical context
+
X
inherited state
fixed
physical readout
Z
physical / chemical dynamics
F
reproductive consequence
differential reproduction changes the distribution of X
(b) Evolutionary history is acquired and expressed separately
HISTORICAL ACQUISITION
historical scaffold H
Δμ
acquired inherited difference
PRESENT EXPRESSION
probe B_P
+
Δμ
$K(B_P)$
R_q
(c) Relational intervention isolates causal effects
same present context and same physical readout
intact μ
B, K_B
$V_B[\mu]$
$E(\mu)$
B, K_B
$V_B[E(\mu)]$
E preserves specified lower-order properties; disrupts target relation
C_D
V_{rel}
compare
(d) Three realizations instantiate the same causal architecture
B
X
Z
F
Model B
segment-dependent physical bias
+
motif–context organization
stochastic local chemistry
adjacency-based reproductive score
Catalytic
reaction-rate field
+
catalyst-dimer relations
four reaction rates
balanced-flux performance
Lattice control
external field
+
inherited binary state
fixed nonlinear readout
nonlinear performance
abstract theoretical control